Policy Iteration:

Effect of discount factor:

- The discount factor controls how much the agent cares about the future traffic conditions
- Gamma between 0.75-0.99 considers long-term traffic impact and more balanced but slower response
- Gamma as 0.50 is balanced between immediate and future rewards and moderate performance.
- Low gamma value 0.25 – 0.10 focuses mainly on immediate queue sizes and fast reactions but unstable
- When gamma is very high, the agent tries to optimize long-term rewards, which may cause slower responses to immediate congestion.
- When gamma is very low, the agent reacts quickly to current traffic but may ignore future congestions patterns

Effect of threshold:

- Large theta(1e-1) = faster convergence but less accurate policy
- Medium (1e-3) = balanced accuracy and computation
- Very small (1e-6) = more precise estimation but longer computation

Traffic congestion problems;

- Efficient polices kept average queue length below 7 vehicles
- The best traffic flow occurred when gamma = 0.10 and theta = 1e-3 with 13 congestion events
- The worst congestion event occurred when gamma = 0.25 and theta = 1e-1 with 60 congestion events
- Efficient policies maintained average queue length around 5-6 vehicles.

Signal switching behaviour:

- Policies with around 65-70 switches slowed better traffic balancing.
- Too few switches (60) caused traffic in one direction

Reward trends:

- Experiments with large congestion events produces more negative rewards
- The best performance occurred when gamma is small

Key findings:

1. Policy iteration learned adaptive traffic control policies

In most experiments average queue length remained between 5-7 vehicles

2. Discount factor affected performance

Gamma = 0.10 produced the best reward (-32.13) while gamma = 0.25 resulted in worst reward (-77.63)

3. Lower theta value improved convergence accuracy

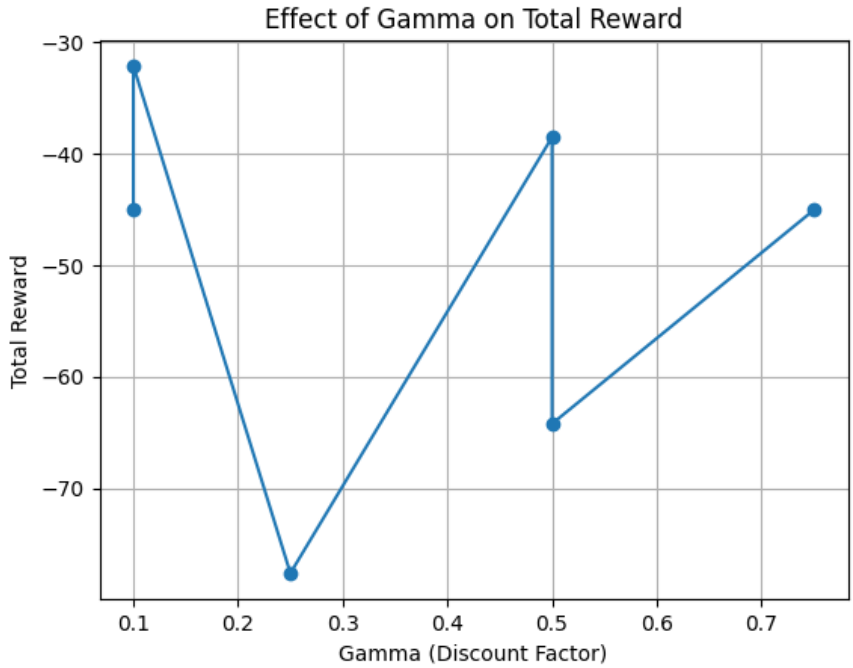
Changing theta from 1e-3 to 1e-6 did not improve performance

4. Balanced signal switching improved traffic flow

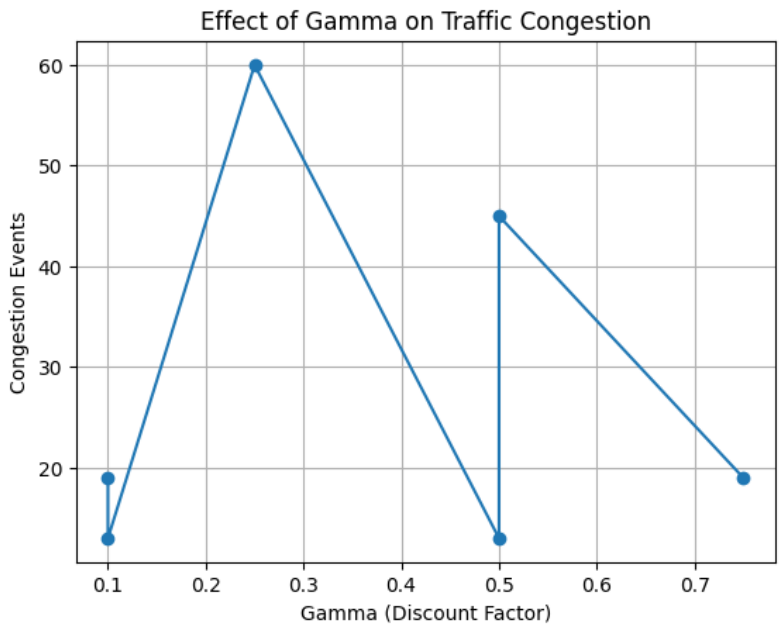
With 65-70 signal switches shared lower congestion compared to runs with fewer switches

5. Policy iteration reduced congestion compared to static traffic signals, keeping the average waiting time vehicles below 7 in most experiments

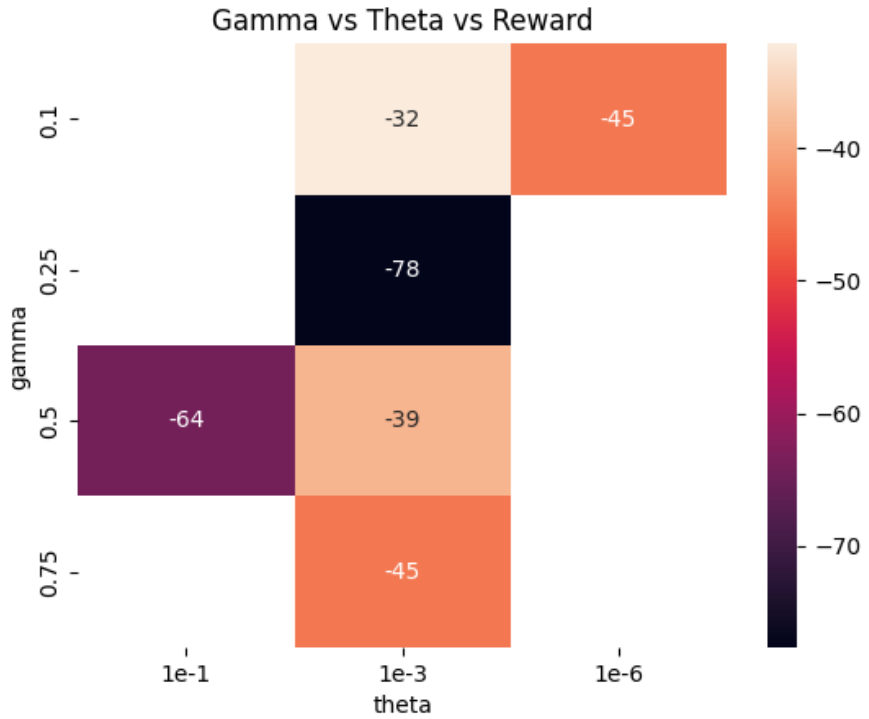
Visualization



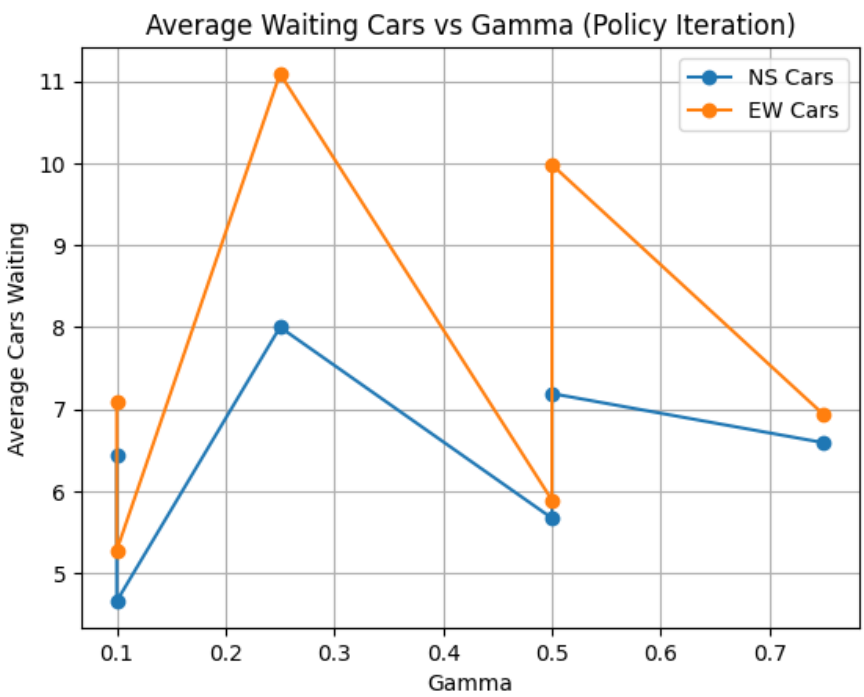
Discount factor decreases, total reward generally increases



Lower gamma value tends to reduce congestion events



Heat map shows that both gamma and theta influence total reward



As the discount factor gamma changes, the average waiting cars in both NS and EW directions vary, indicating that gamma affects how well policy balances traffic flow between two directions

Table comparing all experiments and results:

Experiment	Gamma (γ)	Theta (θ)	Total Reward	Avg NS Cars	Avg EW Cars	Congestion Events
1	0.99	1e-6	-68.33	4.47	13.03	72
2	0.99	1e-3	-39.73	3.62	7.70	26
3	0.75	1e-6	-84.27	10.29	12.59	83
4	0.75	1e-3	-84.27	10.29	12.59	83
5	0.50	1e-1	-38.60	6.45	5.13	15
6	0.50	1e-3	-89.17	11.19	12.56	88
7	0.25	1e-3	-79.13	10.94	10.40	73
8	0.10	1e-6	-79.13	10.74	10.60	70
9	0.10	1e-3	-41.77	6.99	5.54	19

Best configuration:

Gamma = 0.50 and theta = 1e-1, reward = -38.60 with 15 congestions

Worst configuration:

Gamma = 0.50 and theta = 1e-3, reward = -89.17 with 88 congestions